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CS 372
Assignment02

For the first part of this assignment, I tested three different versions of the Vector class: the original version that used a for-loop to copy elements, a version that used the STL `copy()` algorithm, and a version that increased the capacity by a factor of four instead of two. As the number of elements increased, the execution time also increased for all three versions. By making small changes to the code, I observed slight improvements in performance. Increasing the capacity by a factor of four reduced the number of memory reallocations, which helped improve efficiency, although it required more memory to be allocated at once. Overall, both modified versions performed better than the original implementation, with the `copy()` version producing the best timing results.

```
10
20
30
40
50
Program ended with exit code: 0
```

For Question 2, I created an Array class in a separate `.hpp` file. Unlike the Vector class, the Array class has a fixed size that is determined when the object is created and does not change during execution. This allows the data to be stored in a more organized manner and makes it easier to access elements using an index. Overall, the Array class provides a simpler structure for storing data when the size is known in advance.

The Bag class stores items, while the ReceiptBag class stores items and assigns each object a unique receipt number when it is added. This receipt number can later be used to identify and remove a specific item from the collection.

For the ReceiptBag class, I used a similar approach to the Vector implementation by storing data in parallel vectors. When an item needs to be removed, the program searches for the matching receipt number and removes the corresponding item. Testing showed that items were correctly added, tracked, and removed using their assigned receipts.